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MTH623

Final Term (Live Quiz)

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Past Papers for Mids./Finals are also Available

Question 1: Christoffel symbols of first and second kind include derivatives of _____.

Options:

basis vectors

metric tensor ✅

normal vectors

unit vectors

Question 2: The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____.

Options:

quadratic transformations

linear transformations ✅

none of these

non-linear transformations

Question 3: In generalized curvilinear coordinates, the divergence of vector point function

A

i

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=

A

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x

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)

A

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=A

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(x

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) is defined as _____.

Options:

div(Ai) = Ai,i + Γiik

div(Ai) = Ai,i + ΓkiiAk

div(Ai) = Ai,i + ΓiikAk ✓

div(Ai) = Ai,i + ΓiikAk

Question 4: Which of the following represents the covariant derivative of a contravariant vector

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Correct option:

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Question 5: The covariant derivative of an arbitrary tensor is a tensor.

Options:

True ☑

False

Question 6: In generalized curvilinear coordinates, the gradient of scalar function

ϕ

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φ=φ(x

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Options:

∇

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Question 7: For a diagonal metric, which of the following is true?

Options:

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Question 8: Which of the following expressions correctly defines the Christoffel symbols of the first kind?

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Options:

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1

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+g

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Question 9: The Christoffel symbols of second kind are symmetric in _____.

Options:

lower indices ✓

all indices

Question 10: The Christoffel symbols of the first kind satisfy the transformation law of a third-rank tensor.

Options:

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True ☑

False

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